

Linear Regression

egg 2026, Experimental Semantics, day 2

Mojmír Dočekal

2026-07-28

Experimental Linguistics

Two revolutions

- formal linguistics has seen two revolutions since the second half of the 20th century:

1. Chomsky (1957): a cognitive and formal approach to syntax:

the whole goal of science is to replace complex visibles by simple invisibles. That is science. If you are not doing that, then it is something else, it is data organization, flower collection. Sometimes the latter is useful, but it should not be confused with science. If it is science, since Galileo, it is an effort to satisfy Galileo's maxim: nature is simple. If we have not figured it out, it is our problem.

Two revolutions

- formal linguistics has seen two revolutions since the second half of the 20th century:
1. Chomsky (1957): a cognitive and formal approach to syntax:

the whole goal of science is to replace complex visibles by simple invisibles. That is science. If you are not doing that, then it is something else, it is data organization, flower collection. Sometimes the latter is useful, but it should not be confused with science. If it is science, since Galileo, it is an effort to satisfy Galileo's maxim: nature is simple. If we have not figured it out, it is our problem.

2. 1990s: the experimental turn in linguistics (classic textbooks: Baayen (2008), Kruschke (2011))

- perhaps even more important
- analysis of large amounts of data (whether stored in corpora or experimentally collected)
- often these are subtler and more variable data than earlier work in syntax:
- a clear syntactic contrast:

- (1) a. Já jsem nepřišel. (I did not come.)
 b. *Já nejsem přišel. (*I am not come.)

2. 1990s: the experimental turn in linguistics (classic textbooks: Baayen (2008), Kruschke (2011))

- perhaps even more important
- analysis of large amounts of data (whether stored in corpora or experimentally collected)
- often these are subtler and more variable data than earlier work in syntax:
- a clear syntactic contrast:

- (1) a. Já jsem nepřišel. (I did not come.)
 b. *Já nejsem přišel. (*I am not come.)

2. 1990s: the experimental turn in linguistics (classic textbooks: Baayen (2008), Kruschke (2011))
- perhaps even more important
 - analysis of large amounts of data (whether stored in corpora or experimentally collected)
 - often these are subtler and more variable data than earlier work in syntax:
 - a clear syntactic contrast:

- (1) a. Já jsem nepřišel. (I did not come.)
 b. *Já nejsem přišel. (*I am not come.)

2. 1990s: the experimental turn in linguistics (classic textbooks: Baayen (2008), Kruschke (2011))
- perhaps even more important
 - analysis of large amounts of data (whether stored in corpora or experimentally collected)
 - often these are subtler and more variable data than earlier work in syntax:
 - a clear syntactic contrast:

- (1) a. Já jsem nepřišel. (I did not come.)
 b. *Já nejsem přišel. (*I am not come.)

2. 1990s: the experimental turn in linguistics (classic textbooks: Baayen (2008), Kruschke (2011))
- perhaps even more important
 - analysis of large amounts of data (whether stored in corpora or experimentally collected)
 - often these are subtler and more variable data than earlier work in syntax:
 - a clear syntactic contrast:

- (1) a. Já jsem nepřišel. (I did not come.)
 b. *Já nejsem přišel. (*I am not come.)

- a clear example (from NESČ):
 - neutralization of temporal and aspectual distinctions in negated sentences
 - an example from *Bafut* (Cameroon)
 - the contrast between the present perfect and the recent past disappears in negated sentences

- a clear example (from NESČ):
- neutralization of temporal and aspectual distinctions in negated sentences
- an example from *Bafut* (Cameroon)
- the contrast between the present perfect and the recent past disappears in negated sentences

- a clear example (from NESČ):
- neutralization of temporal and aspectual distinctions in negated sentences
- an example from *Bafut* (Cameroon)
- the contrast between the present perfect and the recent past disappears in negated sentences

- a clear example (from NESČ):
- neutralization of temporal and aspectual distinctions in negated sentences
- an example from *Bafut* (Cameroon)
- the contrast between the present perfect and the recent past disappears in negated sentences

- (2)
- a. mbìŋ lòó
rain fell
'It rained/We had rainy weather.'
 - b. mbìŋ lòò me'
rain fell IMPST
'It rained/It had just rained.'
 - c. k̄āā mbìŋ sì l̄òò
NEG rain NEG fell
'It did not rain.'

- neutralization of the aspectual contrast: under negation, we have to use the default form (temporal or aspectual)

- similarly, for Czech it is claimed that (Hajičová, from NESČ):

It is usually said that the interaction of verbal mood and aspect works as follows: in its basic function (prohibition, negative advice, etc.), a perfective verb in the imperative has an imperfective verb as its direct negative counterpart: *Sedni si dopředu!* – *Nesedej si dopředu!* However, this is a broader phenomenon, because a durative verb behaves similarly in relation to an iterative one: *Jedťe zítra do Pardubic!* – *Nejezděte zítra do Pardubic!* Moreover, this is not limited to the imperative, but also applies to various semantically related constructions: *Měli byste jet do Pardubic* – *Neměli byste jezdit do Pardubic*; *Rád by jel do Pardubic* – *Nerad by jezdil do Pardubic*. The boundaries of this phenomenon have not yet been studied systematically (see the survey in Karlík and Nübler (1998)), just as it is not clear to what extent the imperative in the sense of warning is exempt from such variation. This certainly holds, for example, for *Nesedni si na klobouk!*, but it is not entirely clear for the other examples or similar sentences.

- if we want to describe such phenomena, we have to work with more than intuition
- and use tools for handling ambiguous data (signal and noise: https://en.wikipedia.org/wiki/The_Signal_and_the_Noise)
- a historical example: Galton and his study of the inheritance of height
- a similar pattern: intuitively obvious correlation, but many counterexamples

- if we want to describe such phenomena, we have to work with more than intuition
- and use tools for handling ambiguous data (signal and noise: https://en.wikipedia.org/wiki/The_Signal_and_the_Noise)
- a historical example: Galton and his study of the inheritance of height
- a similar pattern: intuitively obvious correlation, but many counterexamples

- if we want to describe such phenomena, we have to work with more than intuition
- and use tools for handling ambiguous data (signal and noise: https://en.wikipedia.org/wiki/The_Signal_and_the_Noise)
- a historical example: Galton and his study of the inheritance of height
- a similar pattern: intuitively obvious correlation, but many counterexamples

- if we want to describe such phenomena, we have to work with more than intuition
- and use tools for handling ambiguous data (signal and noise: https://en.wikipedia.org/wiki/The_Signal_and_the_Noise)
- a historical example: Galton and his study of the inheritance of height
- a similar pattern: intuitively obvious correlation, but many counterexamples

Francis Galton

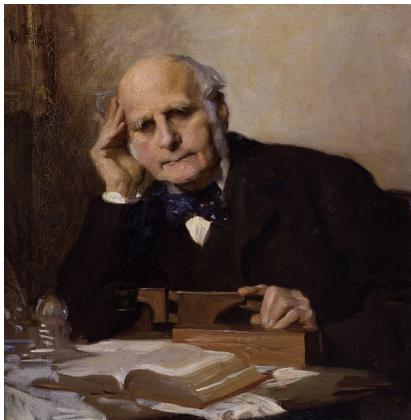


Figure 1: Galton

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- traveler, anthropologist, eugenicist
- foundations of descriptive statistics (median – vox populi) and inferential statistics (linear regression)
- also a meteorologist, statistician (the effectiveness of prayers), and measurer of women's beauty in different parts of England
- a Victorian scholar with a passion for data
- Charles Darwin's cousin
- research question:

(3) If we know the parents' heights, can we predict their children's heights?

- in (frequentist) statistics, a research question is always tested against a null hypothesis (in the classical frequentist framework):

(4) There is no relationship between parents' height and children's height.

- Galton collected data on about 400 parents and their 400 daughters and sons

- in (frequentist) statistics, a research question is always tested against a null hypothesis (in the classical frequentist framework):

(4) There is no relationship between parents' height and children's height.

- Galton collected data on about 400 parents and their 400 daughters and sons

- dataset
 - 1 inch = 2.54 cm
 - today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
 - statistical processing from R Core Team (2021) follows
 - first descriptive statistics
 - the following is output from R (in Czech mathematical terminology, a decimal comma is used)

- dataset
- 1 inch = 2.54 cm
- today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
- statistical processing from R Core Team (2021) follows
- first descriptive statistics
- the following is output from R (in Czech mathematical terminology, a decimal comma is used)

- dataset
- 1 inch = 2.54 cm
- today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
- statistical processing from R Core Team (2021) follows
- first descriptive statistics
- the following is output from R (in Czech mathematical terminology, a decimal comma is used)

- dataset
- 1 inch = 2.54 cm
- today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
- statistical processing from R Core Team (2021) follows
 - first descriptive statistics
 - the following is output from R (in Czech mathematical terminology, a decimal comma is used)

- dataset
- 1 inch = 2.54 cm
- today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
- statistical processing from R Core Team (2021) follows
- first descriptive statistics
- the following is output from R (in Czech mathematical terminology, a decimal comma is used)

- dataset
- 1 inch = 2.54 cm
- today the average height of men and women in the UK is 69 and 63 inches (versus medians of 69.5 and 64 in Galton's sample)
- statistical processing from R Core Team (2021) follows
- first descriptive statistics
- the following is output from R (in Czech mathematical terminology, a decimal comma is used)

```
## [1] "Fathers: počet záznamů, deskriptivní stat."
```

```
## [1] "standard deviation (směrodatná odchylka)"
```

```
## [1] 197
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	62.00	68.00	69.50	69.35	71.00	78.50

```
## [1] 2.622034
```

```
## [1] "Mothers"
```

```
## [1] 197
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	58.00	62.70	64.00	63.98	65.50	70.50

```
## [1] 2.355607
```

```
## [1] "Sons"
```

```
## [1] 465
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      60.00  67.50   69.20   69.23   71.00   79.00
```

```
## [1] 2.631594
```

```
## [1] "Daughters"
```

```
## [1] 433
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      56.00  62.50   64.00   64.11   65.50   70.50
```

```
## [1] 2.37032
```

- quantile: in Czech, lower quantile, median, upper quantile

- important terms: **mean** versus **median**
- standard deviation

```
x <- c(2,4,8,10,100)
mean(x)
```

```
## [1] 24.8
```

```
median(x)
```

```
## [1] 8
```

```
sd(x)
```

```
## [1] 42.15685
```

```
y <- c(2,4,8,10)
sd(y)
```

```
## [1] 3.651484
```

- important terms: **mean** versus **median**
- standard deviation

```
x <- c(2,4,8,10,100)
mean(x)
```

```
## [1] 24.8
```

```
median(x)
```

```
## [1] 8
```

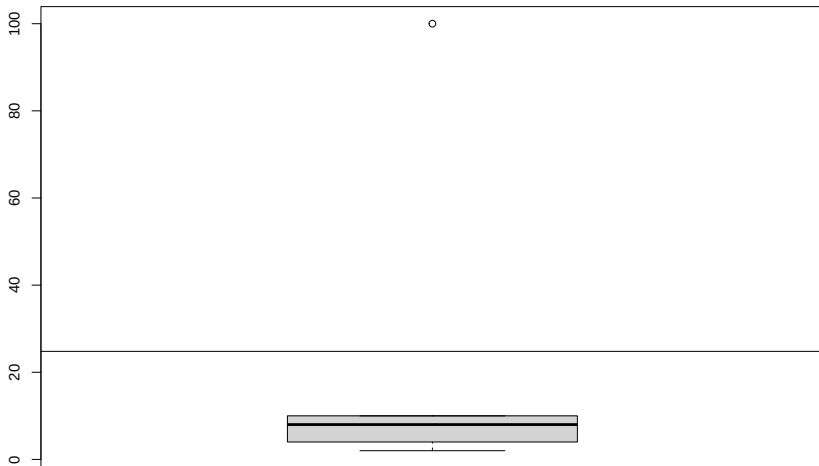
```
sd(x)
```

```
## [1] 42.15685
```

```
y <- c(2,4,8,10)
sd(y)
```

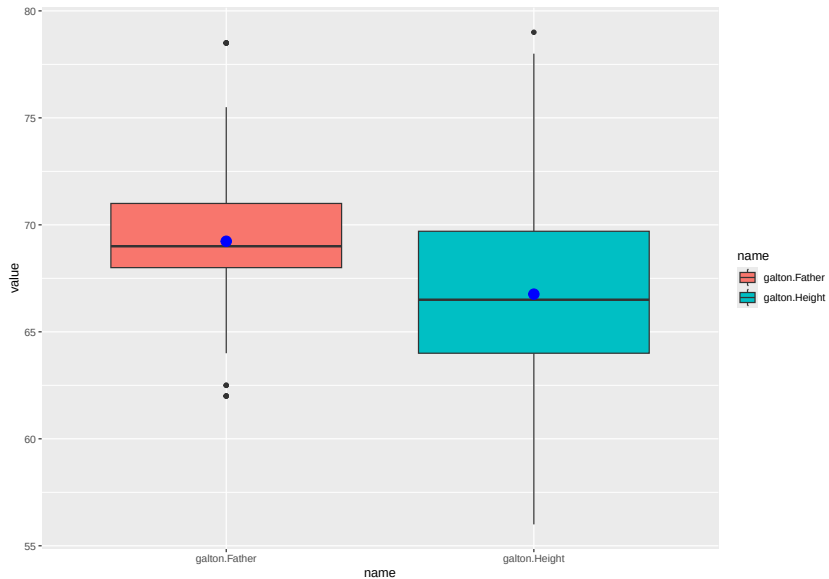
```
## [1] 3.651484
```

```
## [1] "Boxplot (krabicový graf)"
```



- common graphical displays: a boxplot or a scatter plot (correlation diagram)
- descriptive statistics

- common graphical displays: a boxplot or a scatter plot (correlation diagram)
- descriptive statistics



- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)

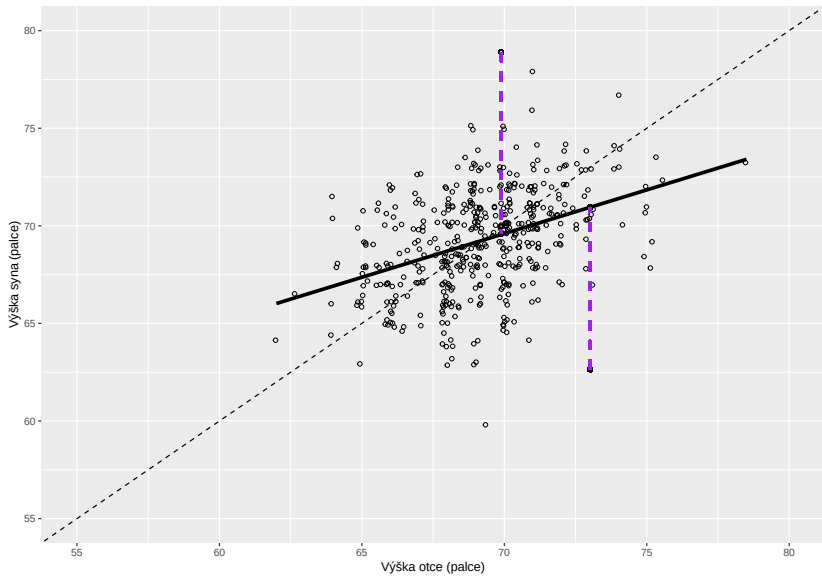
- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)

- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)

- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)

- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)

- the next slide shows a scatter plot of fathers' heights (x) against sons' heights (y)
- jitter is added to distinguish identical values
- dashed line: father height = son height
- thick line (regression line): linear regression, the so-called best fit (best approximation?)
- residual error: the distance of a point from the linear regression line
- according to Spiegelhalter (2019)



- a modern interpretation using the so-called linear model:
- Estimate: how much the son's height changes when the explanatory variable (the father's height) increases by 1 inch, plus the intercept
- t-value: the same as Student's t-test: how far the estimate is from 0, measured in standard errors (more than about 2 in absolute value means a statistically significant effect)
- p-value: the probability under the null hypothesis

- a modern interpretation using the so-called linear model:
- Estimate: how much the son's height changes when the explanatory variable (the father's height) increases by 1 inch, plus the intercept
- t-value: the same as Student's t-test: how far the estimate is from 0, measured in standard errors (more than about 2 in absolute value means a statistically significant effect)
- p-value: the probability under the null hypothesis

- a modern interpretation using the so-called linear model:
- Estimate: how much the son's height changes when the explanatory variable (the father's height) increases by 1 inch, plus the intercept
- t-value: the same as Student's t-test: how far the estimate is from 0, measured in standard errors (more than about 2 in absolute value means a statistically significant effect)
- p-value: the probability under the null hypothesis

- a modern interpretation using the so-called linear model:
- Estimate: how much the son's height changes when the explanatory variable (the father's height) increases by 1 inch, plus the intercept
- t-value: the same as Student's t-test: how far the estimate is from 0, measured in standard errors (more than about 2 in absolute value means a statistically significant effect)
- p-value: the probability under the null hypothesis

```
##
## Call:
## lm(formula = Son ~ FatherS)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.3774 -1.4968  0.0181  1.6375  9.3987
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 38.25891    3.38663   11.30  <2e-16 ***
## FatherS      0.44775    0.04894    9.15  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.424 on 463 degrees of freedom
## Multiple R-squared:  0.1531, Adjusted R-squared:  0.1513
## F-statistic: 83.72 on 1 and 463 DF,  p-value: < 2.2e-16
```

- this is already inferential statistics (inductive reasoning from a sample to the population)
- it is exactly this type of model (though mixed) that we use in linguistics: [Michaela Hulmanová: Asymmetry of cumulative readings with distributive conjunctions: an experimental study](#)

- this is already inferential statistics (inductive reasoning from a sample to the population)
- it is exactly this type of model (though mixed) that we use in linguistics: [Michaela Hulmanová: Asymmetry of cumulative readings with distributive conjunctions: an experimental study](#)

- an illustration of inferential statistics
- <https://www.britannica.com/topic/regression-to-the-mean>
- a serious analysis would have to take Czech data into account: http://www.szu.cz/uploads/documents/obi/CAV/6.CAV_2_Dlouhodobé_zmeny_rustu.pdf

```
MDI <- 194/2.54  
over_median <- MDI - 69.5  
over_median
```

```
## [1] 6.877953
```

```
MDII <- 69.2 + over_median*0.45  
MDII
```

```
## [1] 72.29508
```

```
metric <- MDII*2.54  
metric
```

```
## [1] 183.6295
```

- an illustration of inferential statistics
- <https://www.britannica.com/topic/regression-to-the-mean>
- a serious analysis would have to take Czech data into account: http://www.szu.cz/uploads/documents/obi/CAV/6.CAV_2_Dlouhodobé_zmeny_rustu.pdf

```
MDI <- 194/2.54  
over_median <- MDI - 69.5  
over_median
```

```
## [1] 6.877953  
MDII <- 69.2 + over_median*0.45  
MDII
```

```
## [1] 72.29508  
metric <- MDII*2.54  
metric
```

```
## [1] 183.6295
```

- an illustration of inferential statistics
- <https://www.britannica.com/topic/regression-to-the-mean>
- a serious analysis would have to take Czech data into account: http://www.szu.cz/uploads/documents/obi/CAV/6.CAV_2_Dlouhodobé_zmeny_rustu.pdf

```
MDI <- 194/2.54  
over_median <- MDI - 69.5  
over_median
```

```
## [1] 6.877953  
MDII <- 69.2 + over_median*0.45  
MDII
```

```
## [1] 72.29508  
metric <- MDII*2.54  
metric
```

```
## [1] 183.6295
```

Back to Linguistics

- an example of aspect neutralization in negated imperative sentences
- the research hypothesis is in (5)
- the null hypothesis, by contrast, is in (6):

(5) In Czech imperatives, negated verbs undergo neutralization of the aspectual contrast: the verb is used in the default, i.e. imperfective, aspect.

(6) Negation has no effect on neutralization of the aspectual contrast.

- first step: data

Back to Linguistics

- an example of aspect neutralization in negated imperative sentences
- the research hypothesis is in (5)
- the null hypothesis, by contrast, is in (6):

(5) In Czech imperatives, negated verbs undergo neutralization of the aspectual contrast: the verb is used in the default, i.e. imperfective, aspect.

(6) Negation has no effect on neutralization of the aspectual contrast.

- first step: data

Back to Linguistics

- an example of aspect neutralization in negated imperative sentences
- the research hypothesis is in (5)
- the null hypothesis, by contrast, is in (6):

(5) In Czech imperatives, negated verbs undergo neutralization of the aspectual contrast: the verb is used in the default, i.e. imperfective, aspect.

(6) Negation has no effect on neutralization of the aspectual contrast.

- first step: data

Back to Linguistics

- an example of aspect neutralization in negated imperative sentences
- the research hypothesis is in (5)
- the null hypothesis, by contrast, is in (6):

(5) In Czech imperatives, negated verbs undergo neutralization of the aspectual contrast: the verb is used in the default, i.e. imperfective, aspect.

(6) Negation has no effect on neutralization of the aspectual contrast.

- first step: data

- for this type of data, the [Czech National Corpus](#) is suitable
- searching for words and phrases:
 1. [kernel](#)
 2. [širší než delší](#) (wider than longer)

- for this type of data, the [Czech National Corpus](#) is suitable
- searching for words and phrases:

1. [kernel](#)

2. [širší než delší \(wider than longer\)](#)

- for this type of data, the [Czech National Corpus](#) is suitable
- searching for words and phrases:

1. [kernel](#)

2. [širší než delší \(wider than longer\)](#)

- for this type of data, the [Czech National Corpus](#) is suitable
- searching for words and phrases:
 1. [kernel](#)
 2. [širší než delší](#) (wider than longer)

- searching by morphological tags:

1. baseline (priors):

- a. all imperative imperfectives:

[tag="Vi.....I"]: 104 385

- b. all imperative perfectives: [tag="Vi.....P"]:
165 924

(imperative perfectives dominate over the rest but not in the whole aspectual domain: 11 514 033 (Imperf.) vs. 5 787 695 (Perf.))

- searching by morphological tags:

1. baseline (priors):

- a. all imperative imperfectives:

[tag="Vi.....I"]: 104 385

- b. all imperative perfectives: [tag="Vi.....P"]:
165 924

(imperative perfectives dominate over the rest but not in the whole aspectual domain: 11 514 033 (Imperf.) vs. 5 787 695 (Perf.))

- searching by morphological tags:

1. baseline (priors):

a. all imperative imperfectives:

[tag="Vi.....I"]: 104 385

b. all imperative perfectives: [tag="Vi.....P"]:
165 924

(imperative perfectives dominate over the rest but not in the whole aspectual domain: 11 514 033 (Imperf.) vs. 5 787 695 (Perf.))

- searching by morphological tags:

1. baseline (priors):

a. all imperative imperfectives:

[tag="Vi.....I"]: 104 385

b. all imperative perfectives: [tag="Vi.....P"]:
165 924

(imperative perfectives dominate over the rest but not in the whole aspectual domain: 11 514 033 (Imperf.) vs. 5 787 695 (Perf.))

2. counts of negated perfectives and imperfectives:

a. negated imperfectives: [tag="Vi.....N....I"]: 27
256

b. negated perfectives: [tag="Vi.....N....P"]: 4
567

2. counts of negated perfectives and imperfectives:

a. negated imperfectives: [tag="Vi.....N....I"]: 27
256

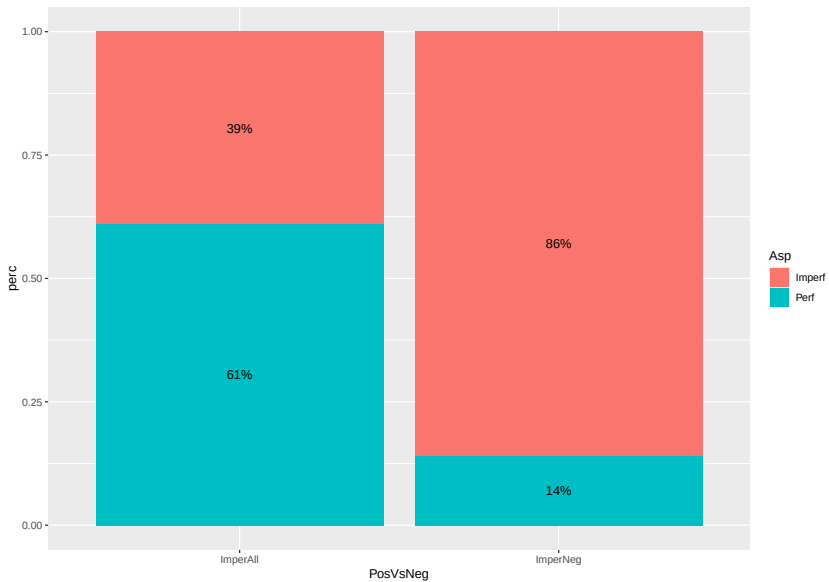
b. negated perfectives: [tag="Vi.....N....P"]: 4
567

2. counts of negated perfectives and imperfectives:

a. negated imperfectives: [tag="Vi.....N....I"]: 27
256

b. negated perfectives: [tag="Vi.....N....P"]: 4
567

- graphs: a boxplot of relative frequencies



- to inferential statistics (from sample to population):
 - for these so-called count data, the most common test is Fisher's exact test or a chi-square test (chi-square distribution)
 - Fisher's exact test:

- to inferential statistics (from sample to population):
- for these so-called count data, the most common test is Fisher's exact test or a chi-square test (chi-square distribution)
- Fisher's exact test:

- to inferential statistics (from sample to population):
- for these so-called count data, the most common test is Fisher's exact test or a chi-square test (chi-square distribution)
- Fisher's exact test:

```
data <- matrix(c(27256, 104385, 4567, 165924), ncol = 2, dimnames = list(c("Imp  
fisher.test(data)
```

```
##  
## Fisher's Exact Test for Count Data  
##  
## data: data  
## p-value < 2.2e-16  
## alternative hypothesis: true odds ratio is not equal to 1  
## 95 percent confidence interval:  
## 9.182734 9.796871  
## sample estimates:  
## odds ratio  
## 9.487635
```


- the probability that the null hypothesis is compatible with such data: $p\text{-value} < 2.2e-16$
- and for the negated imperative it is 9.49 times more likely that the verb will be in the imperfective (compared with all imperatives)
- confidence interval: a 95% interval for how any other random sample in the population would behave: 9.18 – 9.80
- that is, the probability of *nepřečti* in Czech versus *nečti* lies somewhere in this 95% confidence interval

- the probability that the null hypothesis is compatible with such data: $p\text{-value} < 2.2e-16$
- and for the negated imperative it is 9.49 times more likely that the verb will be in the imperfective (compared with all imperatives)
- confidence interval: a 95% interval for how any other random sample in the population would behave: 9.18 – 9.80
- that is, the probability of *nepřečti* in Czech versus *nečti* lies somewhere in this 95% confidence interval

- the probability that the null hypothesis is compatible with such data: $p\text{-value} < 2.2e-16$
- and for the negated imperative it is 9.49 times more likely that the verb will be in the imperfective (compared with all imperatives)
- confidence interval: a 95% interval for how any other random sample in the population would behave: 9.18 – 9.80
- that is, the probability of *nepřečti* in Czech versus *nečti* lies somewhere in this 95% confidence interval

- the probability that the null hypothesis is compatible with such data: $p\text{-value} < 2.2e-16$
- and for the negated imperative it is 9.49 times more likely that the verb will be in the imperfective (compared with all imperatives)
- confidence interval: a 95% interval for how any other random sample in the population would behave: 9.18 – 9.80
- that is, the probability of *nepřečti* in Czech versus *nečti* lies somewhere in this 95% confidence interval

For tomorrow (day 3)

For tomorrow (day 3)

- notebooks
 - either install [Rstudio](#) or use [RstudioCloud](#)
 - depending on your operating system, you may need to install [R](#)
 - and we'll go through intro to [R for Data Science](#)

For tomorrow (day 3)

- notebooks
- either install [Rstudio](#) or use [RstudioCloud](#)
- depending on your operating system, you may need to install [R](#)
- and we'll go through intro to [R for Data Science](#)

For tomorrow (day 3)

- notebooks
- either install [Rstudio](#) or use [RstudioCloud](#)
- depending on your operating system, you may need to install [R](#)
- and we'll go through intro to [R for Data Science](#)

For tomorrow (day 3)

- notebooks
- either install [Rstudio](#) or use [RstudioCloud](#)
- depending on your operating system, you may need to install [R](#)
- and we'll go through intro to [R for Data Science](#)

A Minimal Experiment

- in a crossed-arms task, some people have the right arm on top, ...
- research question: does biological sex affect which one?

(7) Nulová hypotéza: mezi oběma proměnnými není žádný vztah. (Null hypothesis: there is no relationship between the two variables.)

- code: [RstudioCloud](#)

A Minimal Experiment

- in a crossed-arms task, some people have the right arm on top, ...
- research question: does biological sex affect which one?

(7) Nulová hypotéza: mezi oběma proměnnými není žádný vztah. (Null hypothesis: there is no relationship between the two variables.)

- code: [RstudioCloud](#)

A Minimal Experiment

- in a crossed-arms task, some people have the right arm on top, ...
- research question: does biological sex affect which one?

(7) Nulová hypotéza: mezi oběma proměnnými není žádný vztah. (Null hypothesis: there is no relationship between the two variables.)

- code: [RstudioCloud](#)

References i

- Baayen, H. 2008. *Analyzing linguistic data: A practical introduction to statistics using R*. Cambridge University Press.
- Chomsky, Noam. 1957. *Syntactic Structures*. the Hague: Mouton.
- Kruschke, John K. 2011. *Doing Bayesian Data Analysis : A Tutorial with r and BUGS*. Burlington, MA: Academic Press. <http://www.amazon.com/Doing-Bayesian-Data-Analysis-Tutorial/dp/0123814855>.

- R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Spiegelhalter, David. 2019. *The Art of Statistics: Learning from Data*. London: Penguin books.